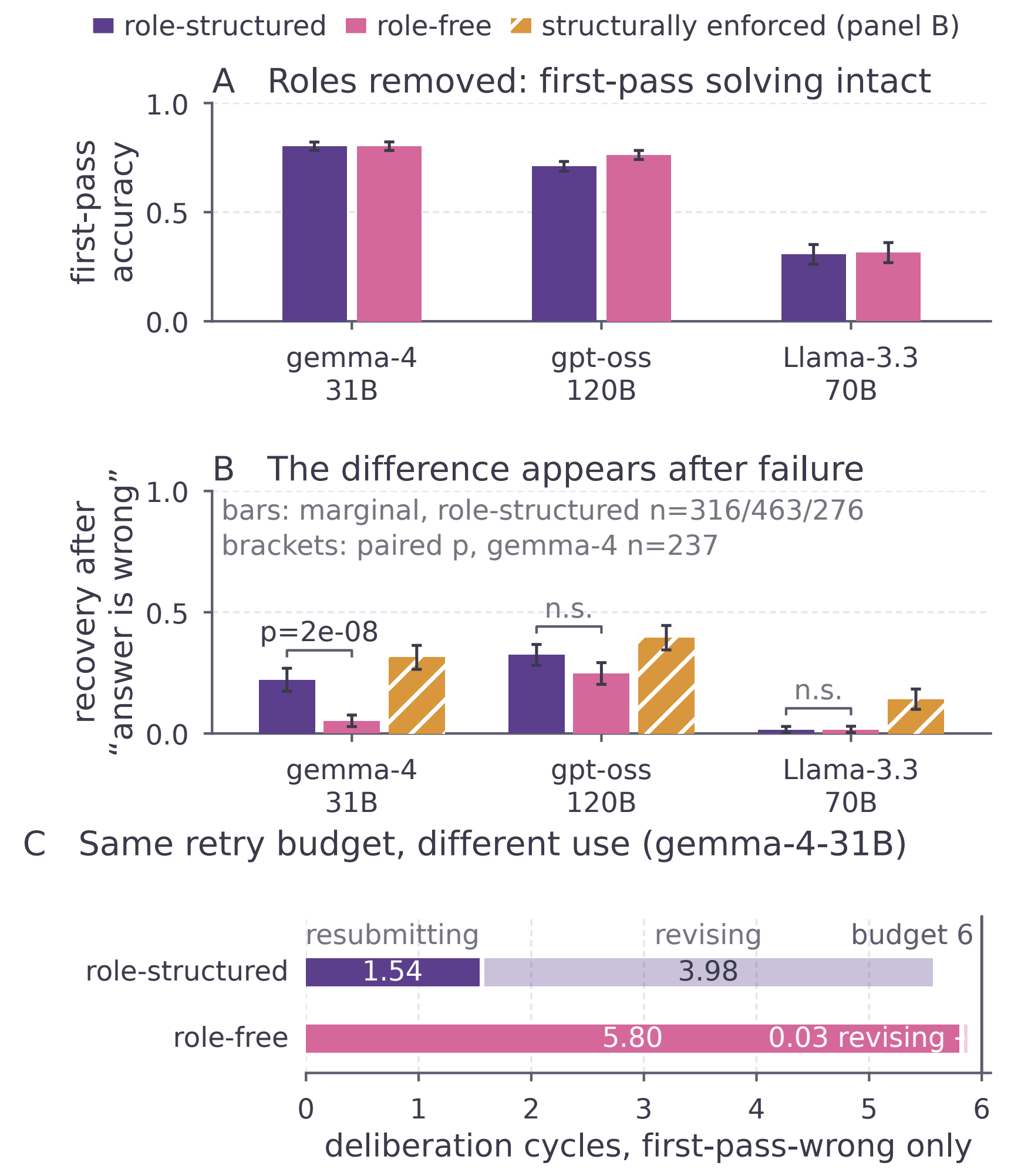
role-structured
role-free
structurally enforced (panel B)
A Roles removed: first-pass solving intact
first-pass accuracy
1.0
0.5
0.0
gemma-4 31B
gpt-oss 120B
Llama-3.3 70B
B The difference appears after failure
recovery after "answer is wrong"
bars: marginal, role-structured n=316/463/276
brackets: paired p, gemma-4 n=237
p=2e-08
n.s.
n.s.
C Same retry budget, different use (gemma-4-31B)
resubmitting
revising
budget 6
role-structured
1.54
3.98
role-free
5.80
0.03 revising
0
1
2
3
4
5
6
deliberation cycles, first-pass-wrong only